

NIFS TOOLS USER MANUAL & TECHNICAL GUIDE



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NIFS TOOLS USER MANUAL & TECHNICAL GUIDE

May 2025 Standards and Practices for National Incident Feature Service Data
Export

**WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES AVIATION OPERATIONS
MULTI MISSION AIRCRAFT (MMA) PROGRAM**

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INTRODUCTION

What is the NIFS Tools?

The NIFS Tools is a specialized ArcGIS Python toolbox developed for Washington State Department of Natural Resources (DNR) Aviation Operations to process National Incident Feature Service (NIFS) data. This tool automates the filtering and export of standardized incident features, including event points, label points, event lines, and event polygons, creating KMZ files optimized for Google Earth and field operations.

NIFS represents the national standard for wildfire incident mapping symbology and feature classification, ensuring consistency across all wildfire management agencies. The NIFS Tools bridges this standardized data with operational field requirements, providing filtered, symbolized exports tailored to specific incidents.

NIFS Integration and Mission Support

The National Incident Feature Service (NIFS) provides standardized geospatial features for wildfire incident management, including:

- **Event Points:** Critical tactical features like helispots, drop points, fire origins, aerial hazards, and UAS operations areas
- **Label Points:** Geographic reference points and text annotations for situational awareness
- **Event Lines:** Linear features including temporary flight restrictions, retardant drops, and aerial hazards
- **Event Polygons:** Area features such as fire perimeters, evacuation zones, and restricted areas
- **Standardized symbology** ensuring consistent interpretation across agencies
- **Real-time incident data integration** from multiple authoritative sources

The NIFS Tools enhances operational effectiveness by:

- **Filtering NIFS data** by specific incident names with wildcard pattern support
- **Applying feature category filters** to focus on aviation-relevant operational features
- **Creating incident-specific KMZ exports** optimized for tactical operations
- **Processing multiple incidents simultaneously** with batch functionality
- **Maintaining data integrity** through comprehensive feature validation
- **Supporting advanced symbology** including rotation and category-based styling

Target Users and Applications

Primary users include:

- DNR Aviation Personnel supporting incident operations
- Incident Management Teams requiring standardized mapping products
- GIS Specialists creating tactical maps and briefing materials

- Air Tactical Group Supervisors coordinating airspace operations
- Division Supervisors planning ground operations
- UAS Operations Coordinators managing unmanned aircraft systems

The tool supports various operational scenarios:

- Incident briefing package creation with current tactical features
- Aviation safety products showing helispots, hazards, and restricted areas
- Resource deployment planning with drop points and staging areas
- Multi-incident campaign support with batch processing capabilities
- Real-time tactical mapping for rapidly evolving incidents

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SECTION I - TOOL OVERVIEW

System Requirements

Software Requirements

- ArcGIS Pro (Version 2.8 or higher recommended)
- ArcGIS Desktop (Version 10.7 or higher, if using ArcMap)
- Python 3.x (included with ArcGIS Pro installation)
- Windows 10 or higher operating system

Network Requirements

- Reliable internet connection for accessing NIFS web services
- Firewall access to services3.arcgis.com domain
- NIFC ArcGIS Online (AGOL) account for NIFS service access
- Minimum 10 Mbps download speed recommended

Hardware Requirements

- **RAM:** Minimum 8GB (16GB recommended for large incident datasets)
- **Storage:** Minimum 1GB available space for temporary files and outputs
- **Processor:** Multi-core processor recommended for batch processing
- **Display:** Minimum 1920x1080 resolution for proper interface viewing

Installation

Step 1: Obtain the Tool

1. Download the NIFS_Tools.pyt file from your DNR system administrator
2. Create a dedicated tools directory (recommended: C:_GIS_Tools)
3. Place the tool file in the designated directory
4. Verify file integrity (file size should be approximately 20KB)

Step 2: Add to ArcGIS Pro

1. Launch ArcGIS Pro
2. Open the Catalog pane (View > Catalog Pane)
3. Navigate to Toolboxes in the catalog tree
4. Right-click Toolboxes and select “Add Toolbox”
5. Browse to the NIFS_Tools.pyt file location
6. Select the file and click “OK”
7. Verify the toolbox appears as “NIFS Tools”

Step 3: Verify Installation

1. Expand the “NIFS Tools” toolbox
2. Confirm “NIFS Incident Export to KML” appears
3. Right-click the tool and select “Properties”
4. Verify the tool description and parameters load correctly
5. Close the Properties dialog

Step 4: Configure Default Paths and Access

1. Verify access to default symbology files:
J:_opsProgram\7-Chief PilotWing (FW)GIS\2025\2025_kml_file_conversionsFiles
2. Verify access to base geodatabase:
J:_opsProgram\7-Chief PilotWing
(FW)GIS\2025\2025_kml_file_conversions\2025_kml_file_comversions.gdb
3. Test network connectivity to NIFS services
4. Confirm NIFC AGOL account access if using service URLs

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SECTION II - USER INTERFACE GUIDE

Parameter Overview

The NIFS Incident Export Tool interface contains seven primary parameters designed for flexible incident data processing:

Layer Selection Parameters

- **Event Point Layer:** Primary tactical features from map contents
- **Label Point Layer:** Reference and annotation features
- **Event Line Layer:** Linear tactical features from map contents
- **Event Polygon Layer:** Area features and boundaries

Processing Configuration

- **Incident Name Filter:** Wildcard-enabled incident name matching
- **Output Folder:** Destination directory for KMZ exports
- **Include Timestamp:** Optional filename timestamping

Parameter Details

Event Point Layer (from Map Contents)

Type: Optional feature layer selection

Source: Active map layers in current ArcGIS Pro project

Supported Geometry: Point features only

Default: None (requires manual selection)

Technical Implementation: This parameter accepts feature layers loaded in the current ArcGIS Pro map, allowing users to work with local data or pre-filtered service layers. The tool processes the selected layer directly from the map contents, maintaining any existing selections or definition queries.

The tool applies automatic feature category filtering for Event Points, limiting exports to operationally relevant features:

- **Aerial Hazard:** Aviation safety concerns and restricted areas
- **Branch Break:** Organizational boundaries for large incidents
- **Division Break:** Geographic operational divisions
- **Fire Origin:** Ignition points and cause determination locations
- **Hot Spot - Spot Fire:** Active fire locations requiring attention
- **UAS Launch and Recovery:** Unmanned aircraft system operations
- **Zone Break:** Specialized operational area divisions

Operational Considerations:

- Load NIFS Event Point layers into your map before running the tool
- Pre-filter layers using definition queries if needed for performance
- Verify layer contains FeatureCategory field for proper filtering
- Consider spatial extent when working with large incident complexes

Label Point Layer

Type: Optional feature layer selection

Source: Map contents or service URL

Default: NIFS Label Point service URL

Supported Geometry: Point features only

Technical Implementation: Label Points provide geographic reference information and text annotations for incident maps. The tool processes these features to maintain spatial reference integrity and text labeling for Google Earth display.

Default Service:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/InternalView_NIFS_2025/FeatureServer/0

Best Practices for Label Points:

- Use default service URL for current national data
- Load custom label layers for specific incident requirements
- Verify text fields are populated for proper Google Earth labeling
- Consider label density for map readability

Event Line Layer (from Map Contents)

Type: Optional feature layer selection

Source: Active map layers in current ArcGIS Pro project

Supported Geometry: Polyline features only

Default: None (requires manual selection)

Filter: Restricted to polyline geometry types

Technical Implementation: Event Lines represent linear tactical features critical to aviation operations. The tool applies automatic feature category filtering for Event Lines, focusing on aviation-relevant linear features:

- **Temporary Flight Restriction:** Airspace restrictions and no-fly zones
- **Aerial Hazard:** Linear hazards affecting aviation operations
- **Retardant Drop:** Fire retardant application lines and patterns

Operational Applications:

- TFR boundaries and restricted airspace
- Retardant drop patterns and effectiveness tracking
- Linear hazards including power lines and communication towers
- Flight path restrictions and safety corridors

Event Polygon Layer

Type: Optional feature layer selection

Source: Map contents or service URL

Default: NIFS Event Polygon service URL

Supported Geometry: Polygon features only

Filter: Restricted to polygon geometry types

Technical Implementation: Event Polygons represent area features critical to incident operations, including fire perimeters, evacuation zones, restricted areas, and operational boundaries. The tool maintains polygon topology and applies appropriate symbology for area feature interpretation.

Default Service:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/InternalView_NIFS_2025/FeatureServer/6

Polygon Feature Applications:

- Fire perimeters and growth projections
- Evacuation and restriction zones
- Operational division boundaries
- Resource staging areas
- Aircraft operational areas

Incident Name Filter

Type: Required text input

Format: Wildcard-enabled incident name matching

Examples: “White”, “Smith Ridge”, “Gold”, “Bear Fire, Pine Creek”

Wildcard Support: Asterisk (*) for pattern matching

Technical Implementation: The tool constructs SQL LIKE queries using the provided incident names, supporting both exact matches and wildcard patterns. Multiple incidents can be processed simultaneously using comma separation.

Pattern Matching Rules:

- **Exact Match:** “Smith Ridge” matches only “Smith Ridge”
- **Prefix Wildcard:** “White*” matches “White River”, “White Rock”, “Whitmore”
- **Multiple Incidents:** “Bear Fire, Pine Creek” processes both incidents
- **Case Insensitive:** All matching performed in uppercase for consistency

SQL Query Construction:

The tool converts wildcard patterns to SQL LIKE syntax:

- “White*” becomes “UPPER(IncidentName) LIKE ‘WHITE%’”
- Comma-separated values create multiple individual queries
- Each incident generates separate output files

Processing Considerations:

- Use wildcards carefully to avoid unintended matches
- Test wildcard patterns with known incident names
- Consider incident naming conventions (e.g., year prefixes)
- Verify incident names exist in NIFS data before processing

Output Folder

Type: Required folder browser

Permissions: Write access required

Default: J:_opsProgram\7-Chief PilotWing (FW)GIS\2025\2025_kml_file_conversions_Files

Recommended: Dedicated folder with adequate space

Technical Implementation: The tool creates all output files in the specified directory using standardized naming conventions. Each incident and feature type combination generates a separate KMZ file with comprehensive logging.

File Management Features:

- Automatic directory creation if path doesn't exist
- Incident-specific file organization
- Feature type separation for easy identification
- Comprehensive processing logs

Storage Considerations:

- Allow minimum 100MB free space per large incident
- Use fast storage devices (SSD recommended) for optimal performance
- Ensure regular backup of output directories
- Consider network storage implications for shared team access

Include Timestamp

Type: Optional boolean checkbox

Default: False (unchecked)

Effect: Adds HHMMSS timestamp to filenames when enabled

Technical Implementation: When enabled, the tool appends the current time (hours, minutes, seconds) to output filenames, enabling multiple processing runs per day without file conflicts.

Filename Examples:

- Without timestamp: 20250525_BearFire_EventPoint.kmz
- With timestamp: 20250525_142315_BearFire_EventPoint.kmz

Timestamp Considerations:

- Enable for multiple daily processing runs
- Useful for tracking processing iterations
- Creates unique files for comparison and archival
- Increases filename length (consider path length limits)

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SECTION III - WORKFLOW INSTRUCTIONS

Basic Workflow

Pre-Execution Checklist

1. Verify ArcGIS Pro is running with current project loaded
2. Confirm internet connectivity and NIFS service accessibility
3. Load required NIFS layers into the current map (Event Points minimum)
4. Ensure adequate disk space in the designated output folder
5. Verify incident names are known and correctly spelled

Step 1: Prepare Map Layers

1. Open ArcGIS Pro with an active map project
2. Add NIFS Event Point layer to map contents: • Catalog > Portal > ArcGIS Online • Search for “NIFS” or use direct service URL • Add Event Point layer to map
3. Optionally add Event Line layer for linear features
4. Verify layers display correctly and contain expected data

Technical Note: The tool works directly with layers in your map contents, allowing you to pre-filter data using definition queries, spatial selections, or custom symbology before processing.

Step 2: Launch the Tool

1. Navigate to Catalog > Toolboxes > NIFS Tools
2. Double-click “NIFS Incident Export to KML” to open interface
3. Allow the tool dialog to fully load and populate parameters

Step 3: Configure Layer Parameters

1. **Event Point Layer:** Select from dropdown (loaded map layers)
2. **Label Point Layer:** Use default service URL or select custom layer
3. **Event Line Layer:** Select from dropdown (loaded map layers)
4. **Event Polygon Layer:** Use default service URL or select custom layer

How the System Works: The tool processes each selected layer type independently, applying incident name filtering to each. Event Points and Event Lines receive additional feature category filtering to focus on operationally relevant features for aviation operations.

Layer Processing Strategy:

- **Event Points:** Filtered by both incident name and feature category
- **Label Points:** Filtered by incident name only
- **Event Lines:** Filtered by both incident name and feature category

- **Event Polygons:** Filtered by incident name only
- Each layer type exports separately for modular use

Step 4: Configure Incident Filtering

1. Enter incident name(s) in the filter field
2. Use wildcards (*) for pattern matching if needed
3. Separate multiple incidents with commas
4. Verify incident names match NIFS database entries

Incident Name Examples:

- Single incident: "Bear Fire"
- Wildcard pattern: "White" (*matches White River, White Rock, etc.*)
- Multiple incidents: "Bear Fire, Pine Creek Fire, Oak Ridge"
- Year-specific: "2025-Bear" (if incidents include year prefixes)

Step 5: Set Output Configuration

1. Browse to select output directory or use default
2. Ensure folder has write permissions and adequate space
3. Choose whether to include timestamps in filenames
4. Consider file organization needs for team sharing

Output Organization Strategy:

- Each incident creates separate files
- Each feature type (Event Point, Label Point, Event Line, Event Polygon) exports individually
- Timestamp option creates unique files for multiple daily runs
- Files use standardized naming for easy identification

Step 6: Execute Processing

1. Review all parameter settings for accuracy
2. Click "Run" to begin processing
3. Monitor Messages window for progress updates
4. Allow processing to complete without interrupting ArcGIS Pro

Expected Processing Times:

- Small incidents (< 100 features per type): 1-3 minutes
- Standard incidents (100-500 features per type): 3-8 minutes
- Large incidents (500+ features per type): 8-15 minutes
- Multiple incident processing: Additional time per incident

Step 7: Verify Results

1. Check output folder for created KMZ files
2. Test files by opening in Google Earth
3. Verify feature counts and symbology display correctly
4. Validate incident name filtering worked as expected

Quality Assurance Checklist:

- All expected output files are present (one per incident/feature type)
- KMZ files open correctly in Google Earth
- Symbology displays according to NIFS standards
- Feature categories are limited to operational types (Event Points and Lines)
- Incident filtering captured correct features

Advanced Configuration

Multi-Incident Batch Processing

Large Campaign Support Configuration

For incidents requiring comprehensive intelligence across multiple fires:

1. **Incident Name Strategy:** • Use wildcard patterns to capture related incidents • Example: “Complex” *captures all complex incidents* • *Process incident families together: “Bear, Pine, Oak”*
2. **Layer Selection:** • Include all four layer types for comprehensive coverage • Use service URLs for most current data • Consider custom layers for specialized requirements
3. **Output Management:** • Enable timestamps for tracking multiple processing runs • Use descriptive output folder names (e.g., “NWCC_Campaign_20250525”) • Plan for large file sets with batch processing

Specialized Feature Filtering

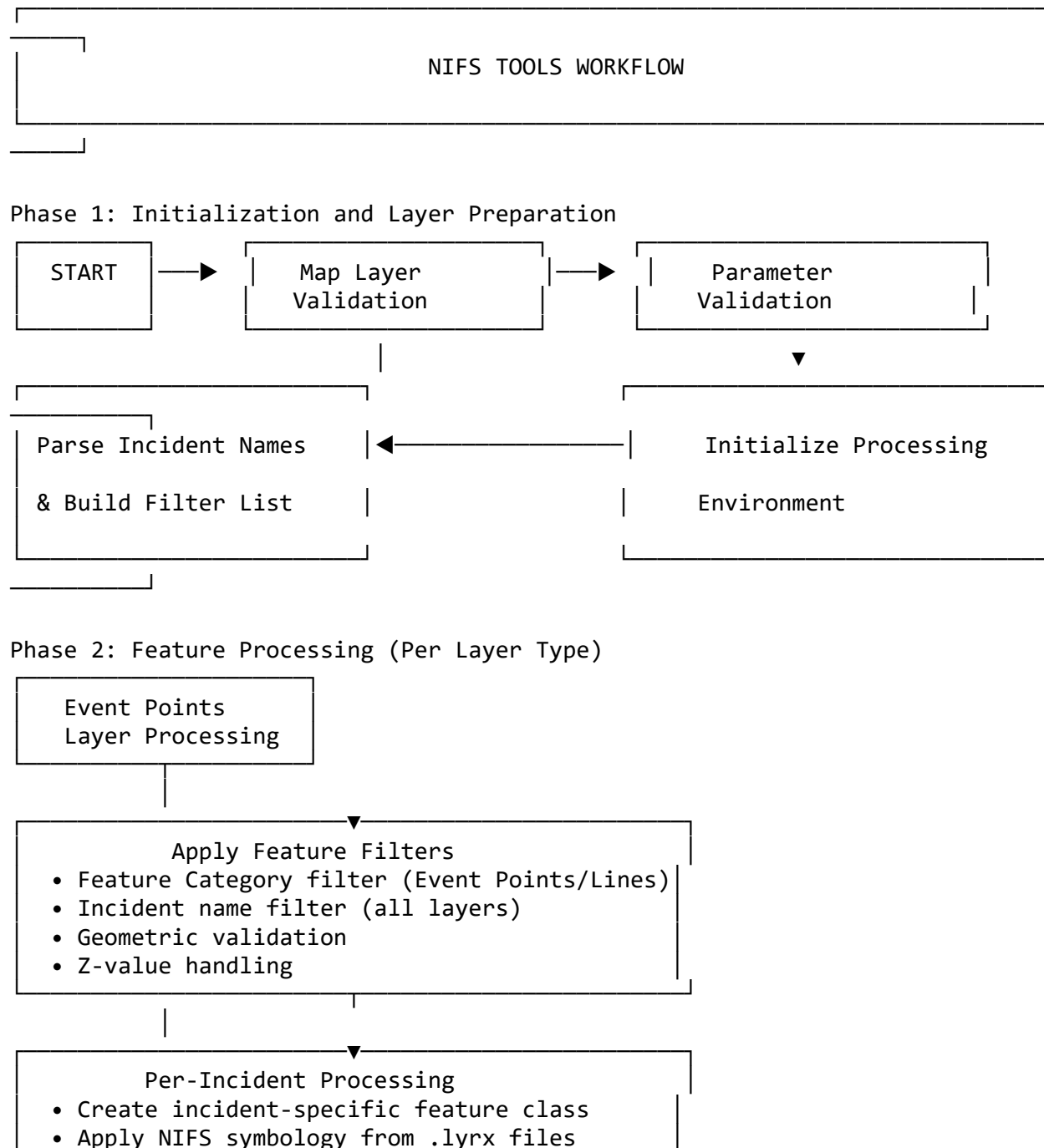
Aviation Operations Focus Configuration

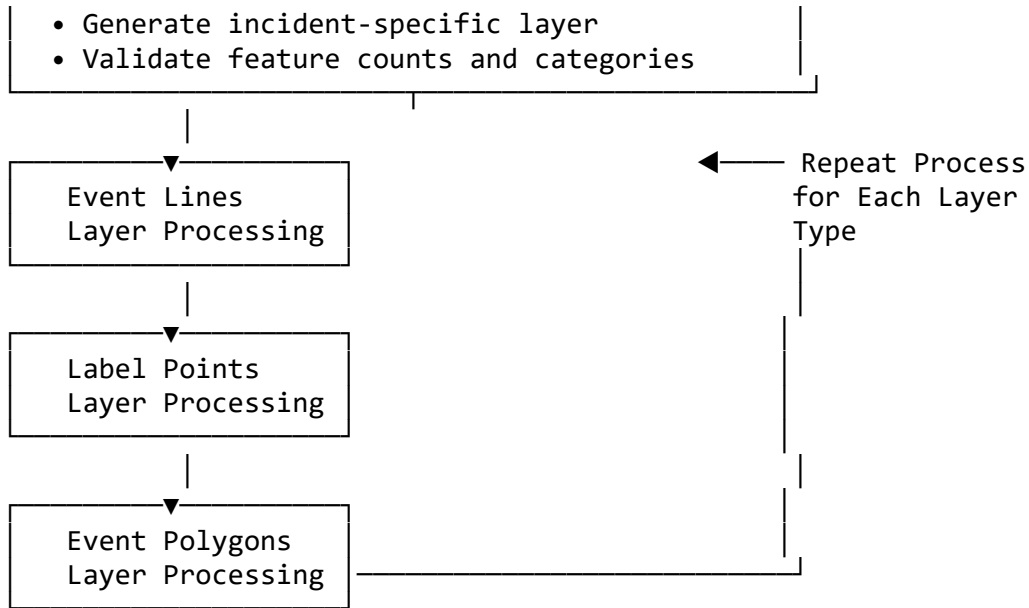
For aviation-specific operations requiring tactical features:

1. **Event Point Categories:** The tool automatically filters to aviation-relevant categories: • **Helispots:** Landing zones and operations areas • **Aerial Hazards:** Safety concerns and restricted areas • **Fire Origin:** Reference points for tactical planning • **UAS Launch and Recovery:** Unmanned aircraft operations • **Hot Spot - Spot Fire:** Active areas requiring attention
2. **Event Line Categories:** The tool automatically filters to operational line features: • **Temporary Flight Restriction:** Airspace restrictions • **Aerial Hazard:** Linear hazards affecting aviation • **Retardant Drop:** Fire suppression application lines

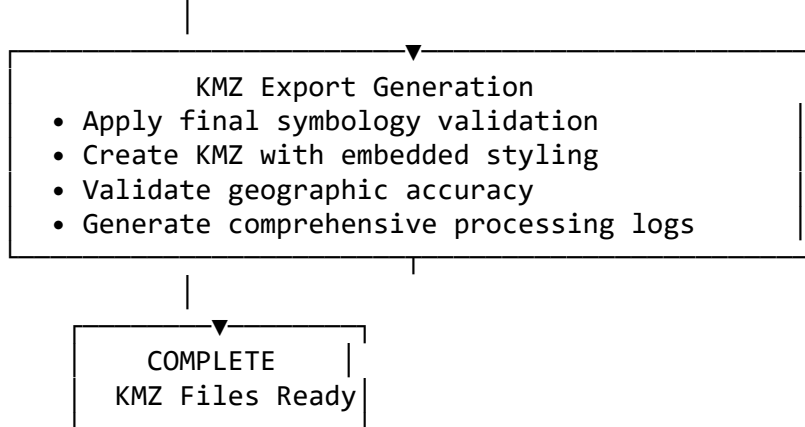
3. **Custom Layer Preparation:** • Pre-filter Event Points using definition queries if needed • Add custom symbology before processing for specialized display • Use spatial selections to focus on specific geographic areas
4. **Quality Control:** • Verify helispot coordinates for accuracy • Confirm drop point accessibility and currency • Validate aerial hazard information completeness

Processing Flow Diagram

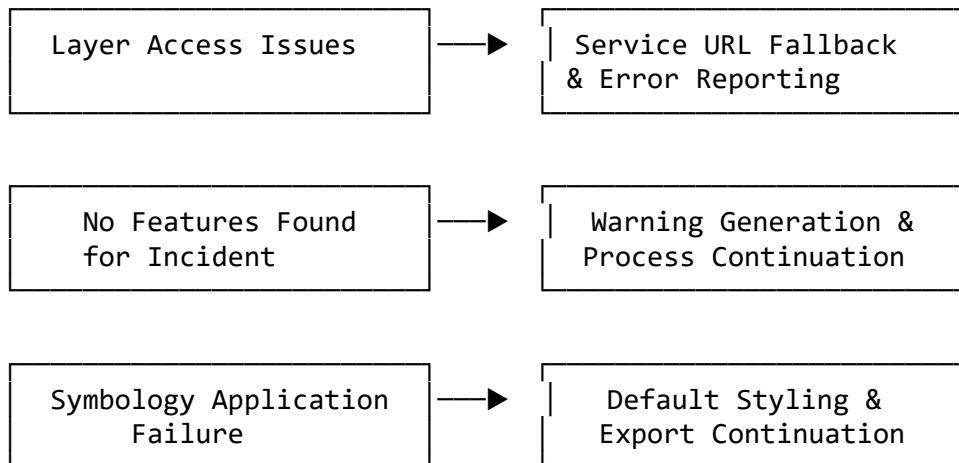




Phase 3: Export and Validation



Error Handling and Quality Control



Key Processing Components:

- **Layer Integration:** Direct map contents layer processing
- **Feature Category Filtering:** Aviation-focused Event Point and Line filtering
- **Incident Name Parsing:** Wildcard and multi-incident support
- **NIFS Symbology:** Standardized NWCG symbol application with rotation support
- **Quality Validation:** Feature count and accuracy verification
- **Batch Processing:** Multi-incident simultaneous processing

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SECTION IV - OUTPUT FILES AND DATA MANAGEMENT

File Naming Conventions

The NIFS Tools implements standardized naming conventions that support efficient file management and clear identification of data currency, incident, and feature type:

Standard Format: YYYYMMDD_[HHMMSS_]INCIDENTNAME_FEATURETYPE.kmz

Component Breakdown:

- **YYYY:** Four-digit year
- **MM:** Two-digit month (01-12)
- **DD:** Two-digit day (01-31)
- **HHMMSS:** Optional timestamp (hours, minutes, seconds) when enabled
- **INCIDENTNAME:** Sanitized incident name (spaces become underscores, special characters removed)
- **FEATURETYPE:** One of EventPoint, LabelPoint, EventLine, or EventPolygon
- **Extension:** Always .kmz for Google Earth compatibility

Example Outputs:

- 20250525_BearFire_EventPoint.kmz - Event points for Bear Fire on May 25, 2025
- 20250525_142315_WhiteRiver_EventPolygon.kmz - Event polygons with timestamp
- 20250525_PineCreek_EventLine.kmz - Event lines for Pine Creek incident
- 20250525_SmithRidge_LabelPoint.kmz - Label points for Smith Ridge incident

Incident Name Sanitization Rules:

- Spaces converted to underscores: “Bear Fire” → “BearFire”
- Wildcards removed: “White*” processes as “White” in filename
- Special characters removed: “Pine-Creek” → “PineCreek”
- Multiple words joined: “Smith Ridge Complex” → “SmithRidgeComplex”

Output Types and Formats

Primary Outputs - KMZ Files

KMZ (Keyhole Markup Language Zipped) files serve as the exclusive output format, optimized for Google Earth integration and field operations:

Content Structure:

- **Spatial Data:** Complete feature geometries with precise coordinates
- **Attribute Information:** All relevant NIFS fields preserved
- **NIFS Symbolology:** Applied NWCG-standard symbols and colors
- **Feature Categories:** Filtered operational features for Event Points and Lines
- **Metadata:** Processing timestamp and source information

Technical Specifications:

- **Coordinate System:** WGS 1984 (EPSG:4326) for universal compatibility
- **Compression:** Optimized for file size while maintaining symbol fidelity
- **Attribute Preservation:** All NIFS standard fields retained
- **Symbol Scaling:** Appropriate for various zoom levels in Google Earth
- **Label Integration:** Text labels preserved for reference points
- **Z-Value Handling:** “Clamped to ground” for KML compatibility

NIFS Feature Type Specifications:

Event Points (EventPoint.kmz)

- **Filtered Categories:** Limited to operational features only
- **Symbol Types:** NIFS-standard point symbols for each category with rotation support
- **Attribute Fields:** FeatureCategory, IncidentName, Angle, plus all NIFS fields
- **Spatial Accuracy:** Survey-grade precision where available
- **Label Display:** Feature category and incident name labeling

Label Points (LabelPoint.kmz)

- **Content:** Geographic reference points and text annotations
- **Symbol Types:** Text-based labels with location markers
- **Attribute Fields:** Label text, coordinates, reference information
- **Display Properties:** Text size and color optimized for Google Earth
- **Spatial Reference:** High-precision coordinate information

Event Lines (EventLine.kmz)

- **Filtered Categories:** Limited to operational linear features
- **Symbol Types:** NIFS-standard line symbols with appropriate styling
- **Attribute Fields:** FeatureCategory, IncidentName, plus all NIFS fields
- **Spatial Accuracy:** Linear precision maintained throughout processing
- **Display Properties:** Line weight and color based on feature category

Event Polygons (EventPolygon.kmz)

- **Content:** Area features including perimeters and zones
- **Symbol Types:** Polygon fills and boundary lines per NIFS standards
- **Attribute Fields:** Area calculations, perimeter data, classification
- **Topology:** Maintained polygon integrity with proper closure
- **Transparency:** Appropriate fill transparency for map overlay

Quality Control and Validation

Automated Processing Validation

The tool implements comprehensive quality control mechanisms throughout processing:

Data Integrity Checks:

- Feature count validation per incident and layer type
- Geometric validity verification for all spatial features
- Attribute completeness assessment for critical NIFS fields
- Coordinate system consistency confirmation
- Feature category validation for Event Points and Lines
- Z-value handling verification

Processing Validation:

- Incident name matching confirmation with feature counts
- Layer accessibility verification before processing
- Symbolology file validation and application confirmation
- Export completion verification with file size checks
- Feature category breakdown reporting

Output File Validation:

- KMZ structure validation and Google Earth compatibility testing
- Feature preservation verification (no data loss during export)
- Symbolology rendering confirmation in target application
- File size reasonableness checks (minimum/maximum thresholds)

Manual Verification Procedures

Post-Processing Quality Assurance:

1. Open each KMZ file in Google Earth to verify display
2. Check feature counts against expected incident size
3. Verify symbolology matches NIFS standards
4. Confirm incident name filtering captured correct features
5. Test file sharing and distribution functionality

Data Accuracy Assessment:

- Cross-reference Event Point locations with known operational features
- Verify Event Line geometry accuracy and completeness
- Verify Label Point text content and positioning accuracy
- Confirm Event Polygon boundaries align with operational requirements
- Test coordinate precision for mission-critical applications

Feature Category Validation:

- Confirm Event Points limited to operational categories only
- Verify Event Lines limited to operational linear features
- Verify aviation-specific features (helispots, TFRs) are included
- Check aerial hazard features for completeness and accuracy
- Validate UAS operations areas are properly identified

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SECTION V - TROUBLESHOOTING AND OPTIMIZATION

Common Issues and Solutions

Network and Service Connectivity

Issue: “Cannot connect to NIFS services”

Symptoms: • Tool fails when using default service URLs

- “Layer not accessible” errors in processing log
- Empty or incomplete data downloads from services

Diagnostic Steps:

1. Test internet connectivity using standard web browser
2. Verify access to <https://services3.arcgis.com> in browser
3. Check NIFC ArcGIS Online account credentials and permissions
4. Confirm NIFS service availability through ArcGIS Online portal
5. Test service URLs directly in ArcGIS Pro

Technical Background: NIFS services require authenticated access through NIFC ArcGIS Online accounts. Connection failures often relate to account permissions or service maintenance rather than network issues.

Solutions:

Service Access Issues: • Verify NIFC AGOL account is active and properly authenticated

- Check service permissions for NIFS internal view access
- Contact NIFC support for service access troubleshooting
- Use alternative service endpoints if available

Network Infrastructure: • Request IT support for firewall configuration review

- Verify proxy server settings don't block ArcGIS service connections
- Test alternative network connections (mobile hotspot for verification)
- Ensure DNS resolution works for Esri service domains

Temporary Workarounds: • Download NIFS data locally and add to map contents

- Use cached/offline NIFS data for time-critical operations
- Process smaller geographic extents to reduce server load
- Coordinate with other users to minimize simultaneous service requests

Data Processing and Feature Issues

Issue: "No features found for incident"

Symptoms: • Tool completes successfully but creates empty KMZ files

- Warning messages about missing features for specified incidents
- Feature counts of zero in processing messages

Root Cause Analysis: • **Incident Name Mismatch:** Spelling variations or case sensitivity

- **Temporal Issues:** Incident not yet in NIFS database or archived
- **Geographic Extent:** Incident outside loaded layer extent
- **Feature Categories:** Event Points or Lines filtered out due to category restrictions

Diagnostic Approach:

1. Verify incident name spelling and capitalization in NIFS database
2. Check if incident exists in source NIFS layers
3. Review incident naming conventions (year prefixes, suffixes)
4. Test with broader wildcard patterns (*Fire, Complex*)
5. Examine Event Point and Line feature categories in source data

Solutions:

Incident Name Troubleshooting: • Use wildcard patterns to capture name variations: "Bear*" instead of "Bear Fire"

- Check for year prefixes: "2025-Bear-Fire" vs "Bear Fire"
- Verify incident names in NIFS database through ArcGIS Online portal
- Use partial matching for complex incident names

Feature Category Issues (Event Points and Lines): • Review available FeatureCategory values in source data

- Understand that only operational categories are processed:

Event Points: Aerial Hazard, Branch Break, Division Break, Fire Origin, Hot Spot - Spot Fire, UAS Launch and Recovery, Zone Break

Event Lines: Temporary Flight Restriction, Aerial Hazard, Retardant Drop

- Contact NIFS administrators if expected categories are missing

Temporal and Geographic Issues: • Verify incident timing (recent incidents may not be in NIFS yet)

- Check map extent covers incident geographic area
 - Use “Zoom to Layer” to verify layer contains expected data
 - Consider incident status (archived incidents may be in different services)
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Symbology and Display Issues

Issue: “Symbology application failed” or “KMZ files display incorrectly in Google Earth”

Symptoms: • Features appear with default symbols instead of NIFS symbology

- Rotation not applied to Event Point symbols
- Features appear in wrong locations
- Text labels missing or improperly formatted

Symbology Troubleshooting:

Symbology File Issues: • Confirm symbology files exist at specified paths: - Event_Point.lyrx - Label_Point.lyrx - Event_Line.lyrx - Event_Polygon.lyrx • Verify .lyrx file compatibility with current ArcGIS Pro version

- Test symbology application with sample data before processing
- Check for NIFS symbology updates and version compatibility

Advanced Symbology Features: • Verify Angle field exists for Event Point rotation

- Confirm FeatureCategory field exists for category-based styling
- Check field mapping in symbology files matches data schema
- Test fallback symbology options when advanced features fail

Display Problems: • Verify coordinate system handling (WGS 1984 expected)

- Check for coordinate transformation errors in source data
- Test with known reference locations for accuracy validation
- Compare coordinates between ArcGIS Pro and Google Earth display

Performance Optimization

Processing Speed Enhancement

Optimal Configuration Settings:

Layer Management: • Load only required layers into map contents

- Use definition queries to pre-filter large datasets
- Remove unnecessary layers to reduce memory usage
- Consider spatial extent limitations for large regions

Incident Processing: • Process related incidents together using wildcards

- Limit simultaneous incident processing (< 10 incidents)
- Use specific incident names rather than broad wildcards when possible
- Schedule processing during off-peak service hours

System Resources: • Run ArcGIS Pro with administrator privileges

- Close resource-intensive applications during processing
- Use local storage for temporary processing files
- Ensure adequate RAM (8GB minimum, 16GB recommended)

Advanced Performance Tuning:

Data Management Strategies: • Pre-download NIFS layers for offline processing when possible

- Use local geodatabase storage for frequently accessed data
- Create spatial indexes on frequently queried fields
- Consider data caching for repeated processing runs

Processing Optimization: • Enable multiprocessing for batch incident processing

- Use SSD storage for temporary file operations
- Optimize network bandwidth during service access
- Monitor system resources during processing

Best Practices

Data Preparation Best Practices

1. **Load required layers** into your map before running the tool
2. **Check field names** match expected schema (IncidentName, FeatureCategory, Angle)
3. **Verify coordinate systems** are appropriate for your area of interest
4. **Test connectivity** to NIFS services before critical operations

Incident Name Filtering Best Practices

1. **Use wildcards strategically** - Fire* captures all incidents starting with “Fire”
2. **Multiple incidents** - separate with commas: Fire A, Fire B, Mountain*
3. **Test filters** with a few incidents before processing large batches
4. **Document naming conventions** for your organization

Output Management Best Practices

1. **Organize by date** - tool automatically includes date in filename
2. **Use timestamps** for multiple runs on same day
3. **Monitor disk space** in output directory
4. **Backup important exports** for archival purposes

Quality Control Best Practices

1. **Review tool messages** for warnings about missing data
2. **Check feature counts** in messages to verify expected results
3. **Test KMZ files** in target applications before distribution
4. **Validate feature categories** match operational requirements
5. **Cross-reference** with known incident information for accuracy

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SECTION VI - TECHNICAL REFERENCE

Code Architecture

Tool Structure Overview

The NIFS Tools is implemented as a Python toolbox (.pyt) containing a single tool class designed for robust processing of NIFS incident data:

Primary Components:

- **Toolbox Class:** Container for tool organization and metadata
- **NIFSIncidentExportTool Class:** Core processing logic and parameter management
- **Parameter Management:** Dynamic parameter configuration and validation
- **Processing Engine:** Feature filtering, symbology application, and export functionality
- **Error Handling:** Comprehensive error management and user feedback

Key Technical Features:

Layer Processing Architecture: • Direct integration with ArcGIS Pro map contents

- Support for both local layers and web services
- Automatic geometry type validation and handling
- Z-value preservation and “clamped to ground” export
- Feature-class-to-feature-class processing for geometry stability

Symbology System: • Advanced symbology application with rotation support (Event Points)

- Category-based styling using FeatureCategory field
- Fallback mechanisms for symbology application failures
- Integration with NIFS-standard .lyrx symbology files

Filtering Logic: • Dual-level filtering: incident name + feature category

- SQL LIKE query construction for wildcard pattern matching
- Case-insensitive incident name processing
- Multi-incident batch processing with individual file generation

NIFS Data Structure

Required Field Schema

The tool expects NIFS-compliant data structure with the following critical fields:

Universal Fields (All Feature Types): • **IncidentName** (Text): Primary incident identifier

• **Shape** (Geometry): Spatial feature representation

• **ObjectID** (Integer): Unique feature identifier

Event Points Additional Fields: • **FeatureCategory** (Text): Point feature classification

• **Angle** (Double): Rotation angle for symbol display

• Additional NIFS attributes as defined by national standards

Event Lines Additional Fields: • **FeatureCategory** (Text): Line feature classification

• Additional NIFS attributes as defined by national standards

Supported Feature Categories:

Event Points: • Aerial Hazard

• Branch Break

• Division Break

• Fire Origin

• Hot Spot - Spot Fire

• UAS Launch and Recovery

• Zone Break

Event Lines: • Temporary Flight Restriction

• Aerial Hazard

• Retardant Drop

Coordinate System Requirements

• **Input:** Any valid spatial reference system

• **Processing:** Native coordinate system maintained during processing

• **Output:** WGS 1984 Geographic Coordinate System (EPSG:4326) for KMZ compatibility

Maintenance and Updates

Regular Maintenance Tasks

Monthly: • Verify NIFS service URL accessibility and currency

• Test symbology file compatibility with current ArcGIS Pro version

• Review processing logs for recurring issues or performance degradation

• Validate default file paths and directory accessibility

Quarterly: • Update symbology files to latest NIFS standards if available

• Review feature category filter lists for additions or modifications

- Test tool functionality with latest ArcGIS Pro updates
- Backup and archive processing logs and configuration files

Annually:

- Update default service URLs for current year NIFS data
- Review and update file path references for annual directory structures
- Update version information and documentation
- Conduct comprehensive testing with annual NIFS data updates

Code Modification Guidelines

Safe Modification Areas:

- Default parameter values (service URLs, file paths)
- Feature category filter lists
- Output file naming conventions
- Logging and message text

Areas Requiring Careful Testing:

- Symbology application logic
- Geometry processing and coordinate system handling
- Error handling and fallback mechanisms
- Parameter validation and dependency management

Areas Requiring Expert Knowledge:

- Core processing algorithms
- ArcPy integration and spatial operations
- KMZ export functionality
- Multi-incident batch processing logic

Version Control and Documentation:

- Maintain version history with detailed change documentation
- Test all modifications with sample data before deployment
- Document any changes to default configurations
- Coordinate updates with DNR Aviation Operations team

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GLOSSARY

ArcGIS Online (AGOL) - Cloud-based geographic information system platform hosting NIFS services

Event Lines - Linear tactical features in NIFS including TFRs, retardant drops, and aerial hazards

Event Points - Point-based tactical features in NIFS including helispots, drop points, and hazards

Event Polygons - Area-based features in NIFS including fire perimeters and operational boundaries

Feature Category - Classification field in NIFS data defining the operational type of each feature

KMZ - Keyhole Markup Language Zipped format for Google Earth compatibility

Label Points - Reference and annotation features providing geographic context

NIFC - National Interagency Fire Center, the authoritative source for NIFS data

NIFS - National Incident Feature Service, standardized wildfire incident mapping data

NWCG - National Wildfire Coordinating Group, establishing wildfire symbology standards

Temporary Flight Restriction (TFR) - Designated airspace restrictions during incident operations

UAS - Unmanned Aircraft System, including drones and remote-piloted aircraft

Wildcard Pattern - Text search pattern using asterisk (*) for flexible incident name matching

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END OF DOCUMENT

This manual represents the complete documentation for the NIFS Tools Version 1.0. For updates and additional resources, please contact the DNR Aviation GIS Team.

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